

Candidate ID _____ Score _____

The 29th International Biology Olympiad National Team Selection Camp in 2018

Plant practical exercises

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

I. Materials and Experimental Equipment:

A. Experimental plant materials: (Wild-type Col-0 and mutant b were used only) 1. Arabian			quantity
mustard (wild-type Col-0 and mutant b) were grown under white light for 2 weeks before being transplanted into (red light + blue light)			One plate (4 kinds)
It grows in light (light intensity 100 $\mu\text{mol}/\text{m}^2 \text{ s}$) for a period of time.			Seedlings)/person
B. Experimental Equipment:	Quantity C: Drug solution: 1 unit		quantity
1. Optical/Dissecting Microscope	each of equilibrium solutions (30 mM KCl, 10 mM MES-KOH, pH 6.15)		10 ml/person
2. Electronic eyepiece	1 equilibration solution (containing 10 μM MeJA)		10 ml/person
3. Notebook computer	1 equilibration solution (containing 50 μM ABA)		10 ml/person
4. Scissors/Ruler/Computer	1 distilled water each		10 ml/person
5. 3cc dropper	3		
6. Tweezers (with fine pointed tips)	2		
7. Glass slide	6 pieces		
8. Coverslip	12 pieces		
9. Absorbent paper	1 box		
10. Single-sided blade	1 piece		
11. Oil-based ballpoint pen (fine tip)	1 piece		
12. Washi tape	Volume 1		
13. Matte tape	Volume 1		
14. Separated petri dishes (already labeled with exam numbers)	4 plates		
15. Timer	1		

Precautions:

1. Please confirm that your candidate number is correct; if it is incorrect, please raise your hand and ask the teaching assistant for assistance.
2. Once the medicines and equipment on the table are used up, they will not be replenished.
3. Please use public instruments in accordance with the instructions.
4. This exam paper (including the cover and the question paper) has a total of 5 pages and must be returned in its entirety when submitting the exam.
5. The total time allotted for this exam is 90 minutes. Please answer the questions on this paper. Answers may be written on the back of the questions, but please...

Please indicate and label the question with a number.

Part 1 investigates the effects of light intensity and hormones on the stomata of Arabidopsis:

This practical exercise utilizes white light, red light, blue light, and different hormones to treat Arabidopsis (wild-type Col-0).

And after a period of time, mutant b) was observed under a microscope to understand the state of stomatal opening on the lower epidermis of the leaves.

The effects of light and hormones on the stomata of Arabidopsis.

II. Experimental Methods, Results and Discussion:

Experiment 1: Observation of stomata by directly tearing the leaf epidermis.

1. Take wild-type Col-0 mustard (grown under red + blue light) grown in a light source with a total light intensity of 100 $\mu\text{mol}/\text{m}^2 \text{ s}$.

And one leaf from each of mutant b) seedlings was used to observe the stomatal size.

2. Sample preparation

Please cut a piece of washi tape approximately 5.5 cm long and 2.5 cm wide, sticky side up, and fix it to the table. Take some mustard greens (wild mustard greens).

Leaves of type Col-0 and mutant b) were prepared by flattening the epidermis onto masking tape and then applying matte tape.

Press gently to adhere the lower skin layer, forming a sandwich-like structure, then carefully peel off the matte tape to reveal the plant's lower skin layer.

It will be torn off along with the matte tape.

Depending on the different experimental treatments described below, wild-type and mutant strain b can be marked on the corner of the matte tape.

To distinguish them. strains Please prepare three sets of matte tape containing the lower epidermis of wild-type and mutant)!!!

3. Hormone treatment

Please add the equilibration solution (30 mM KCl, 10 mM MES-KOH, pH 6.15) to the separate culture dishes.

Three groups of matte adhesive tape containing the lower epidermis of wild-type and mutant strains were immersed in balanced solutions in separate culture dishes.

The plant was placed in a liquid and kept in a plant growth chamber with the original light intensity for **15** minutes.

Please add the following A, B, and C solutions to the separated culture dishes:

A. Control group: Equilibrium solution (containing 30 mM KCl, 10 mM MES-KOH, pH 6.15)

B. MeJA group: Equilibrium solution (equilibrium solution containing 10 μM MeJA and methylated jasmonic acid)

C. ABA group: Equilibrium solution (equilibrium solution containing 50 μM ABA and absorptive acid)

After reacting for 15 minutes, transfer the tape from the lower epidermis of the wild-type and mutant strains into separate ABC-separated culture dishes.

Inside: Immerse the matte tape on the lower epidermis in the solution of the separated culture dish, and continue under the original light conditions.

The growth chamber reaction lasted for **20** minutes .

Take the sample containing the plant's lower epidermis along with the matte tape and place it on a glass slide. Use a micropipette to drop the previously treated sample onto the slide.

Add ABC solution, then cover with a coverslip.

4. Observe the size of the pores in each sample using an optical microscope, and record and archive the observations using an electron eyepiece. (6 points)

Both the file name and file title should be labeled: Examiner Number - Image Number. (e.g., F1-1 indicates Examiner Number F1 - Wild-type Control Group)

sample	wild-type mutant b	
control group	1	2
10 μM MeJA	3	4
50 μM ABA	5	6

5. Measure the stomata size of each treated "wild-type" sample using an optical microscope. Measure the length of the long axis (L; parallel to the long axis of the guard cells) and the short axis (S; perpendicular to the long axis of the guard cells). Repeat the measurement for three different stomata to obtain the average value. Note: Under a 40x objective lens, each graduation of the eyepiece micrometer is... The unit is "U"; the size of the pores is expressed as L x S (U), and is recorded in Table 2. Please evaluate the size of the pores to one decimal place.

Results and Discussion:

1. Based on the observed stomatal shape and size of each sample, depict (i.e., draw) in the table below. (2 points per cell , **12 points in total**) Wild

sample	type	Mutant b
control group		
10 μ M Table		
50 μ M ABA		

2. Stomatal size (U) of wild-type samples after each treatment (**15 points**)

Light intensity (μ mol/ m ² s) Control group	100		
average			
10 μ M Table			
average			
50 μ M ABA			
average			

3. Following on from question 1, based on the observed stomatal size, try to deduce which mutant mutant b is most likely to be.

Mutant b: _____ (3 points) Explanation: _____ (5 points)

Does mutant b respond to MeJA and ABA in the same way as the wild type? (3 points) (Check one)

Same: _____, or different: _____;

Please explain why? (5 points)

4. Following on from question 1, if the control group samples (wild-type Col-0 and mutant b) are placed in the dark for 4 hours, compare the stomatal size with that of the samples in question 1. (3 points) (Check one)

Mutant b: _____ increases; _____ decreases; _____ remains unchanged.

Please explain why? (5 points)

5. In this experiment, please explain the reason for the red light treatment when observing the stomata size. (8 points)

reason:

Part Two: Morphological and Anatomical Comparison of Arabic Mustard Leaves and True Leaves:

To compare the differences between cotyledons and true leaves in different Arabica mustard strains (wild-type Col-0 and mutant b), various characteristics were observed using optical or dissecting microscopes, and a comparison table is presented below. (The cotyledons and true leaves are then compared.)

Take photos of each leaf to record its characteristics, and then list them out one by one. In addition to textual descriptions of the characteristics, you can also add drawings to supplement them.

Ming. (Add your own feature column) (Total **35 points**)

1. Take photos of the cotyledons and true leaves separately. (10 **points**)

Please use an electronic eyepiece to take photos and archive them. You may also add text descriptions to the photos.

Please note: Both the photo and the archive name must be: Exam Number. Otherwise, no points will be awarded. Use

an optical microscope (4x objective lens) to take photos and point out the differences

between the two. 2. Comparison Table (**25 points**)

feature	Wild type	Mutant b